

ITSOLERA Pvt. Ltd.
Machine Learning Internship Project Proposal

Project Title: Content-Based Movie Recommendation System Using Machine Learning

1. Introduction

Online streaming platforms like Netflix and Amazon Prime recommend movies based on user interests and viewing history. These recommendations help users discover content that matches their preferences. This project focuses on building a **Content-Based Movie Recommendation System** that suggests movies similar to a selected movie. The system uses movie features such as genre and description to identify and recommend related movies using Machine Learning and Natural Language Processing techniques.

2. Objectives

The objectives of this project are:

- To understand how recommendation systems work
- To extract meaningful features from movie text data
- To build a similarity-based movie recommendation system
- To recommend movies based on content similarity

3. Literature Review

Recommendation systems are widely used in entertainment platforms to improve user experience. Content-based recommendation approaches focus on analyzing item features rather than user behavior.

Previous studies show that text feature extraction techniques such as TF-IDF and cosine similarity are effective in finding similarities between movies based on genre and descriptions. Content-based systems are simple, interpretable, and effective for personalized recommendations.

4. Methodology

4.1. Data Collection

We will use a publicly available **movie dataset** containing information such as movie titles, genres, and descriptions. This dataset will serve as the basis for generating movie recommendations.

4.2. Data Preprocessing

- **Text Cleaning:** Convert text to lowercase and remove punctuation and special characters.
- **Stopword Removal:** Remove common words that do not contribute to movie similarity.

- **Tokenization:** Split movie descriptions into individual words.
- **Lemmatization:** Convert words into their base form for better matching.

4.3. Feature Extraction and Recommendation

- **Text Vectorization:** Use TF-IDF Vectorizer to convert movie descriptions and genres into numerical features.
- **Similarity Measurement:** Apply Cosine Similarity to measure similarity between movies.
- **Recommendation Logic:** Recommend top similar movies based on similarity scores.

4.4. Model Evaluation

- **Recommendation Quality:** Evaluate whether recommended movies are relevant and similar.
- **Manual Validation:** Test recommendations using different input movies.
- **Result Analysis:** Analyze similarity scores to ensure meaningful recommendations.

5. Expected Outcomes

- A content-based movie recommendation system.
- List of recommended movies based on selected movie.
- Understanding of similarity-based recommendation logic.
- Practical exposure to recommendation systems used in real-world platforms.

6. Conclusion

This project aims to develop a content-based movie recommendation system using Machine Learning techniques. By analyzing movie features such as genre and description, the system provides relevant movie suggestions. The project helps students understand recommendation logic and gain hands-on experience with real-world recommendation engines.